Python program project

PROJECT PROPOSAL

Development of an Offline Android Tourist Guide Application for Chilimo Forest Using Python and Kivy

A Project Proposal Submitted to

[AMBO UNIVERSITY STEM CENTER]

In Partial Fulfillment of the Requirements for the Completion of the Python Programming Course

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Approval Sheet

This project proposal entitled "Development of an Offline Android Tourist Guide Application for Chilimo Forest Using Python and Kivy" has been reviewed and approved as a partial fulfillment of the requirements for the completion of the Python Programming Course.

Instructors' Name: _____

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Acknowledgement

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Abstract

Chilimo Forest is one of Ethiopia's most important natural forests and is known for its rich biodiversity, indigenous tree species, wildlife, and ecological importance.

Although many people visit the forest for tourism, education, and research, obtaining complete and organized information during a visit can be difficult because internet connectivity may be limited or unavailable in there area.

This project proposes the development of an Offline Android Tourist Guide Application for Chilimo Forest Using Python and Kivy. The application will provide users with detailed information about the forest, including its history, location, flora, fauna, tourist attractions, roads, hiking trails, conservation activities, visitor guidelines, image gallery, and offline map. All information will be stored locally in an SQLite database so that the application can function without an internet connection.

The application will be developed using

Python programming language with the Kivy framework and will follow the Software Development Life Cycle (SDLC) methodology. The completed application is expected to provide visitors, students, and researchers with a simple, reliable, and user-friendly way to access information about Chilimo Forest while also demonstrating practical Python programming skills.

Keywords: Chilimo Forest, Python, Kivy, SQLite, Android Application, Offline Tourist Guide.

CHAPTER ONE

INTRODUCTION

1.1 Background of the Study

Technology has transformed the way people access information and explore tourist destinations. Mobile applications have become an important tool for providing information because they are portable, easy to use, and can offer quick access to valuable resources. In recent years, Android applications have been widely used in tourism to provide visitors with information about historical sites, natural attractions, cultural heritage, transportation, and visitor services.

Chilimo Forest is one of the most important natural forests in Ethiopia. It is located in the Oromia Region, approximately 90 kilometers west of Finfinne. The forest is one of the country's remaining dry Afro-montane forests and is recognized for its rich biodiversity, indigenous tree species, wildlife, bird species, and beautiful natural environment. It also plays an important role in environmental conservation, climate regulation, water resource protection, and scientific research.

Every year, tourists, students, researchers, and environmental experts visit Chilimo

Forest. However, many people who have a strong interest in visiting Chilimo Forest, which is a tourist attraction, are unable to visit it due to various challenges, such as security problems, road/accessibility difficulties, poor internet connectivity, and a shortage of tourist guides.In addition, even when people try to visit the forest using digital means, they cannot find complete information about the forest in one place. Moreover, the information they do find is not always reliable and trustworthy.

To address this problem, this project proposes the development of an Offline Android Tourist Guide Application for Chilimo Forest Using Python and Kivy. The application will provide users with organized and comprehensive information about the forest without requiring an internet connection. It will include information on the forest's history, location, flora, fauna, tourist attractions, roads, hiking trails, conservation activities, visitor guidelines, photographs, and an offline map. The application aims to improve visitors' experiences while also supporting education, research, and environmental awareness.

1.2 Statement of the Problem

Although Chilimo Forest is one of Ethiopia's important natural attractions, there is no dedicated offline Android application that provides comprehensive information about the forest in a single platform. Most available information is accessible only through websites or printed materials, which may not be available to visitors while they are inside the forest.

As a result, visitors often spend considerable time searching for information from different sources. This limits their understanding of the forest and reduces the quality of their experience. Students and researchers also face challenges in collecting organized and reliable information.

Therefore, there is a need to develop an offline Android application that stores information locally and enables users to access detailed information about Chilimo Forest at any time without internet connectivity.

1.3 General Objective

The general objective of this project is to develop an Offline Android Tourist Guide Application for Chilimo Forest using Python and Kivy that provides comprehensive information to visitors without requiring an internet connection.

1.4 Specific Objectives

The specific objectives of this project are:

To collect accurate and reliable information about Chilimo Forest.

To design a user-friendly Android application.

To develop the application using Python and the Kivy framework.

To create an SQLite database for offline data storage.

To provide information about the history, flora, fauna, tourist attractions, roads, hiking trails, and conservation activities of Chilimo Forest.

To include an offline map and image gallery.

To implement a search feature for easy access to information.

To test and evaluate the performance of the application.

1.5 Research Questions

The project seeks to answer the following questions:

How can an offline Android application improve access to information about Chilimo Forest?

What information should be included to support visitors, students, and researchers?

How can Python and Kivy be used to develop an effective offline mobile application?

How can SQLite be used to manage information without an internet connection?

1.6 Significance of the Study

This project is expected to provide several benefits.

It will help tourists obtain reliable information while visiting Chilimo Forest.

It will support students and researchers by providing organized educational information.

It will promote awareness of environmental conservation.

It will demonstrate the practical application of Python programming in Android application development.

It will improve the developer's programming, database, and mobile application development skills.

1.7 Scope of the Study

The project focuses on developing an offline Android application for Chilimo Forest. The application will provide information about the forest's history, location, flora, fauna, roads, hiking trails, tourist attractions, conservation activities, visitor guidelines, image gallery, and offline map. All information will be stored locally using SQLite. Online booking, live GPS navigation, and real-time internet services are outside the scope of this project.

1.8 Limitations of the Study

The project may face limitations such as limited development time, limited financial resources, and limited availability of updated information and photographs. In addition, the application will initially cover only Chilimo Forest and will not include other tourist destinations.

1.9 Organization of the Proposal

This proposal is organized into three chapters. Chapter One presents the introduction, problem statement, objectives, significance, scope, and limitations of the study. Chapter Two reviews related literature and previous studies. Chapter Three describes the methodology that will be used to develop the proposed system.

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

This chapter presents a review of literature related to the proposed project. It discusses information about Chilimo Forest, tourism information systems, offline mobile applications, Python programming language, Kivy framework, SQLite database, and related works. The purpose of this chapter is to understand existing knowledge and identify the gap that the proposed application will address.

2.2 Overview of Chilimo Forest

Chilimo Forest is one of the remaining

natural forests in Ethiopia and is located in the Oromia Region, west of Finfinnee. It is classified as a dry Afro-montane forest and contains important indigenous plant species. The forest has ecological, economic, and social importance for the surrounding communities and the country.

The forest provides many environmental services, including protection of biodiversity, conservation of soil and water resources, carbon storage, and support for local livelihoods. Due to its natural beauty and biological diversity, Chilimo Forest has potential for ecotourism, education, and scientific research.

2.3 Biodiversity of Chilimo Forest

Biodiversity refers to the variety of living organisms found in an ecosystem. Chilimo Forest contains different types of plants, animals, and bird species.

2.3.1 Flora (Plant Diversity)

The forest contains various indigenous tree species and other plants. These plants are important for ecological balance, traditional medicine, and environmental protection.

The application will provide information about:

Plant names

Descriptions

Uses

Images

Conservation status

2.3.2 Fauna (Animal Diversity)

Chilimo Forest provides habitat for different wildlife species. These animals contribute to the ecological importance of the forest.

The application will include:

Animal names

Descriptions

Habitat information

Images

2.3.3 Bird Species

Forests are important habitats for many bird species. Information about birds will help visitors and students understand the biological value of Chilimo Forest.

2.4 Tourism and Information Technology

Tourism is an important sector that

benefits from modern information technology. Mobile applications help tourists access information about destinations, attractions, maps, and services.

A tourism guide application can improve visitor experiences by providing organized information in an easy-to-use format. However, many tourism applications depend on internet connectivity, which creates challenges in remote areas.

2.5 Offline Mobile Applications

An offline mobile application is a software application that can operate without an internet connection. It stores necessary information inside the device using local storage or databases.

Offline applications are useful in areas where:

Internet coverage is limited.

Users need information during travel.

Quick access to information is required.

For Chilimo Forest, an offline application is suitable because visitors may not always have access to mobile networks while exploring the forest.

2.6 Android Mobile Platform

Android is one of the most widely used mobile operating systems in the world. It provides a flexible platform for developing applications for smartphones and tablets.

Android applications can provide:

User-friendly interfaces

Multimedia support

Location-based services

Offline data storage

The proposed project focuses on developing an Android application because of the popularity and accessibility of Android devices.

2.7 Python Programming Language

Python is a high-level, interpreted programming language known for its simplicity and readability. It supports different programming concepts, including procedural programming and object-oriented programming.

Python provides features such as:

Variables and data types

Conditional statements

Loops

Functions

Classes and objects

File handling

Database connectivity

In this project, Python will be used as the main programming language to develop the application.

2.8 Kivy Framework

Kivy is an open-source Python framework used for developing graphical user interface applications. It allows developers to create applications that can run on different platforms, including Android.

Kivy provides components such as:

Buttons

Labels

Images

Text fields

Screen navigation

The framework is selected because it allows Python developers to create mobile applications without learning a completely different programming language.

2.9 SQLite Database

SQLite is a lightweight relational database management system. It stores data in a single file and does not require a separate server.

SQLite is suitable for this project because:

It works offline.

It requires low storage space.

It is easy to integrate with Python.

It provides fast data access.

The database will store information about:

Forest history

Plants

Animals

Tourist attractions

Roads

Images

Visitor guidelines

2.10 Related Works

Different tourism applications have been developed to provide information about tourist destinations. Many of these systems provide information about locations, attractions, and services.

However, most existing tourism applications require internet access and focus on general tourist areas. Few applications focus on providing detailed offline information about specific natural forests in Ethiopia.

2.11 Research Gap

Although Chilimo Forest has significant ecological and tourism value, there is no dedicated offline Android application that provides complete information about the forest.

Existing information is mainly available through websites, documents, and research papers, which may not be convenient for visitors during forest exploration.

Therefore, this project aims to fill this gap by developing an offline Android tourist guide application that provides organized information about Chilimo Forest using Python, Kivy, and SQLite.

2.12 Chapter Summary

This chapter reviewed literature related to Chilimo Forest, biodiversity, tourism applications, offline systems, Python programming, Kivy framework, and SQLite databases. The review identified the need for an offline mobile application that provides detailed information about Chilimo Forest.

CHAPTER THREE

METHODOLOGY

3.1 Introduction

This chapter describes the methods and procedures that will be used to develop the proposed Offline Android Tourist Guide Application for Chilimo Forest Using Python and Kivy. It explains the development approach, data collection methods, system requirements, system design, database design, implementation, and testing methods.

3.2 Development Approach

The project will use the Software Development Life Cycle (SDLC) approach. SDLC provides a systematic process for developing software from planning to implementation and evaluation.

The main phases of the development process are:

3.2.1 Requirement Analysis

In this phase, the requirements of the application will be identified. Information will be collected about the needs of tourists, students, researchers, and other users.

The main requirements include:

Information about Chilimo Forest

History and location

Flora and fauna information

Tourist attractions

Roads and hiking routes

Offline map

Image gallery

Search functionality

Visitor guidelines

3.2.2 System Design

In this phase, the structure and design of the application will be prepared.

The design includes:

User Interface Design

Home screen

Menu navigation

Information display pages

Image viewing pages

Search page

Database Design

Designing tables to store forest information.

System Flow Design

Designing how users interact with the application.

3.2.3 Implementation

The application will be developed using Python programming language and Kivy framework.

The main technologies are:

ComponentTechnologyProgramming LanguagePythonMobile FrameworkKivyDatabaseSQLiteDevelopment EnvironmentVisual Studio Code/PyCharmAndroid Package ToolBuildozer

Python concepts that will be applied include:

Variables and data types

Conditional statements

Loops

Functions

Object-Oriented Programming

File handling

Exception handling

Database connectivity

3.2.4 Testing

Testing will be performed after development to ensure that the application works correctly.

The testing methods include:

Functional Testing

Checking whether all features work properly.

Interface Testing

Checking whether the application is easy to use.

Database Testing

Checking whether information is stored and retrieved correctly.

Performance Testing

Checking application speed and reliability.

3.2.5 Evaluation

After testing, the application will be evaluated by allowing users to use the system and provide feedback. The feedback will be used to improve the application.

3.3 Data Collection Methods

The information required for the application will be collected using the following methods:

3.3.1 Document Review

Different sources such as:

Research papers

Books

Websites

Reports

Educational materials

will be reviewed to collect accurate information about Chilimo Forest.

3.3.2 Observation

Observation will help identify important locations, tourist attractions, and information needed by visitors.

3.3.3 Interview

If possible, interviews will be conducted with local residents, forest experts, or visitors to collect additional information.

3.4 System Requirements

3.4.1 Functional Requirements

The application should:

Display information about Chilimo Forest.

Show history and location.

Provide flora and fauna information.

Display images.

Provide offline map information.

Allow users to search information.

Provide visitor guidelines.

Work without internet connection.

3.4.2 Non-Functional Requirements

The application should:

Be easy to use.

Have a simple interface.

Work efficiently on Android phones.

Require low storage space.

Provide fast access to information.

3.5 System Architecture

The proposed system will have three main layers:

3.5.1 User Interface Layer

This layer allows users to interact with the application. It contains screens, buttons, images, and information pages developed using Kivy.

3.5.2 Application Logic Layer

This layer controls the operations of the application, including searching, navigation, and processing user requests.

3.5.3 Database Layer

This layer stores and manages information using SQLite. It allows the application to work offline.

3.6 Database Design

The application will use SQLite as an offline database.

The main tables include:

Forest Information Table

FieldDescriptionForest_IDUnique identificationTitleInformation titleDescriptionForest descriptionLocationForest locationHistoryHistorical information

Flora Table

FieldDescriptionPlant_IDUnique identificationPlant_NameName of plantDescriptionPlant informationImagePlant image

Fauna Table

FieldDescriptionAnimal_IDUnique identificationAnimal_NameName of animalDescriptionAnimal informationImageAnimal image

Attraction Table

FieldDescriptionAttraction_IDUnique identificationNameAttraction nameDescriptionDetailsLocationPlace information

3.7 Expected Results

At the end of the project, a functional offline Android application will be developed.

The application will:

Provide detailed information about Chilimo Forest.

Help visitors explore the forest easily.

Work without internet connection.

Demonstrate Python programming skills.

Support environmental education and tourism awareness.

3.8 Chapter Summary

This chapter explained the methodology that will be used to develop the proposed system. It discussed the development approach, data collection methods, system requirements, design, database structure, implementation tools, testing methods, and expected outcomes.